Théo **Schifferli**

Machine Learning Scientist at BIIE

theo.schifferli@immune.engineering | theoschifferli.ch | theoschiff | theoschiff

Education

École Polytechnique Fédérale de Lausanne (EPFL)
Lausanne, Switzerland

M.Sc. in Data Science – Computational Biology Minor — *Feb. 2024 - July 2026*

- Thesis on CAPO: Compute-Aware Automated Protein-Model Optimization, supervised by Thomas Bikias and Mary-Anne Hartley. Grade: 6/6.
- Notable projects:
  - Multimeditron: Developed at LiGHT Lab, focusing on a modular multimodal medical LLM trained from scratch with MoE and embedding-projection fusion.
  - Logprob-reward GRPO: Explored confidence-based rewards to elicit LLM reasoning across math and poetry tasks.

B.Sc. in Computer Science — *Sep. 2020 – Jan. 2024*

- Thesis on Multimodal Large Language Models for food recognition, supervised by Prof. Marcel Salathé. Grade: 6/6.

Research Experience

Machine Learning Scientist
Lausanne, Switzerland

Botnar Institute of Immune Engineering (BIIE) — *Oct. 2026 – present*

- Researching novel architectures and agentic systems for biology, with efficient training and rigorous evaluation.

Master's Thesis Researcher
Lausanne, Switzerland

Botnar Institute of Immune Engineering, AI Initiative — *Jan. 2026 – Jul. 2026*

- Developed CAPO, a compute-aware agentic system for end-to-end protein-model optimization across three tasks (host-range MCC 0.826 vs. 0.743 for an expert transformer; 57.4% less GPU time than a coding agent).
- Trained a transformer for RBD–ACE2 binding prediction on 8.8M interactions across 15 species (MCC 0.830).

Research Intern
Lausanne, Switzerland

Botnar Institute of Immune Engineering, AI Initiative — *Sep. 2025 – Dec. 2025 (20%)*

- Built a GraphRAG pipeline (entity graph, Personalized PageRank) over the immunology corpus, benchmarked against production RAG.

Summer Research Intern, Regina Barzilay Group
Cambridge, MA, USA

MIT Computer Science and Artificial Intelligence Laboratory (CSAIL) — *Aug. 2022 – Sep. 2022*

- Fine-tuned sequence models for pathogenicity prediction across 70 ClinVar cancer genes, achieving 0.875 AUC.

Publications & Preprints

CAPO: Compute-Aware Automated Protein-Model Optimization
2026

Théo Schifferli, Julián García Pardiñas, Edward B. Irvine, Mary-Anne Hartley, Thomas Bikias.

bioRxiv preprint. ***NeurIPS 2026*** *Agentic AI for Biological Discovery Workshop.*

Predicting viral host range for reverse zoonosis surveillance
2026

Edward B. Irvine, Thomas Bikias, **Théo Schifferli**, Leon Plath, Jiami Han, NingNing Chen, Berk Sahin, Lester Frei, Joseph M. Taft, Sai T. Reddy.

bioRxiv preprint.

Awards & Honours

Best Poster Award
OxML 2026 Health & Bio Track; awarded for CAPO research — *2026*

2nd Place – IEEE SSCI 2025
Explainable AI in Personalised Cancer Medicine Challenge — *2025*

Teaching Experience

Student Assistant in Modern NLP — *Lausanne, Switzerland*

EPFL – Course taught by Antoine Bosselut — *Feb. 2025 - Jul. 2025*

- Mentored master's students in weekly labs and supervised and graded 12 final projects for CS-552: Modern NLP.

Student Assistant in Theory of Computation — *Lausanne, Switzerland*

EPFL – Course taught by Mika Göös — *Feb. – Jul. 2023; Feb – Jul 2024*

- Supported students with course exercises during weekly two-hour sessions for CS-251: Theory of Computation.

Private Tutor — *Geneva, Switzerland*

Mathematics, Physics & Programming — *2020 – 2022*

- Provided individual tutoring to final-year high-school students in mathematics, physics and programming, helping them prepare for the Swiss Matura examinations.

Technical Skills

Programming	Python, C/C++, Java, Scala, C#, R, SQL, Assembly, VHDL
ML Stack	PyTorch, NumPy, SciPy, Pandas, Jupyter, Hugging Face Transformers, Linux Command Line, Git
Bio/Protein	Biopython, ESM, FASTA, AlphaFold, Boltz, ColabFold
Languages	French (Native Speaker), English (Professional Proficiency), German (B2 Level), Spanish (B1 Level)
Interests	Modern Art, Cooking, Long Distance Running, Music Production/Guitar, Skiing (Freestyle/Freeride), Triathlon

Additional Work Experience

Summer Intern, Data Intelligence & Digitalisation — *Geneva, Switzerland*

Union Bancaire Privée (UBP) — *Aug. 2024 – Sep. 2024*

- Built Python, Flask and R Shiny tools to automate multi-timeframe trading-signal analysis and reporting.

Extracurricular Activity

Not-So-Fast Run Club — *Geneva, Switzerland*

Co-Founder and President — *May 2024 – present*

- Co-founded and lead a non-profit Geneva running association with inclusive weekly runs, community events, partnerships with brands (including On, Saucony and Vitamin Well) and 1,100+ Strava members.